

LHUSON & ASSOCIATES

REPORT

REVIEW

Lal Lal Wind Farm Environmental Noise Assessments

CLIENT:

DST Legal

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L HUSON & ASSOCIATES PTY LTD

[Consulting Scientists in Acoustics](#)

PO Box 290
WOODEND
VIC 3442

Tel. 0416 143 716

Email: office@huson.com.au

REVIEW

Lal Lal Wind Farm Environmental Noise Assessments

EXECUTIVE SUMMARY

L Huson & Associates Pty Ltd (LHA) has reviewed a number of reports prepared by Arup, SLR and Marshall Day Acoustics for the Lal Lal Wind Farm. In addition, an analysis of raw data collected by SLR has been completed.

SLR completed a sound monitoring survey at dwelling J14aa from 30 October 2020 to 1 December 2020 in response to complaints about noise from the residents. The Elaine section of the Lal Lal Wind Farm is the closest to dwelling J14aa but at the time of the survey the Elaine section was not operating at full capacity due to imposed generation constraints. Accordingly, data collected during this monitoring period cannot be used to assess compliance with permit conditions but may be used to identify the cause of noise distress reported by the residents of dwelling J14aa at that time.

Background noise data has not been gathered at J14aa. It has been proposed by SLR that background measurements from another dwelling (K15aa) approximately 2.3km away be used in lieu. This report finds that Background noise data from K15aa must not be used and notes that a noise compliance report prepared by SLR has incorrectly interpreted wind speed data at K15aa in its assessment. This error has not been identified by the Auditor of these reports.

The SLR compliance assessment report for Elaine contains a number of errors and has failed to assess amplitude modulation. This has not been raised as a concern by the Auditor despite the lack of sufficient attended observational surveys to meet the requirements of the NCTP that could support the lack of objective assessment.

The NCTP has used an incorrect and unproven technique to assess amplitude modulation, contrary to the requirement of NZS6808:2010.

Amplitude modulation has been identified in data recorded by SLR in their complaint investigation that was not considered by SLR. Amplitude modulation of an 80 Hz tone has been identified which could be the cause of noise disturbance at J14aa.

The level of the 80 Hz tone recorded outside J14aa was at levels exceeding 30 dB(A) that is considered sufficient to exceed the DEFRA NAN-R-45 proposed indoor criterion for assessing low frequency noise. Attenuation from outdoors to indoors at 80 Hz is small (<5 dB) which would exceed the recommended indoor sound level limit of 18 dB(A). Under these circumstances low frequency noise complaints are expected.

Sound from the Lal Lal Wind Farm wind turbines contain tonal components that were not considered in pre-construction noise modeling (Sonus Pre-construction Noise Assessment S5464C6, April 2018). In hindsight, if tonal components were expected and an appropriate penalty included in noise models then it is unlikely that the wind farm would have been approved in its current form.

Compliance noise measurements by SLR demonstrate that Background noise levels listed to set target noise limits in the NCTP are too high.

TABLE OF CONTENTS

EXECUTIVE SUMMARY	1
TABLE OF CONTENTS.....	2
INTRODUCTION	3
DETAILED REVIEW	3
Arup Audit	3
Attended subjective listening surveys.....	5
SLR complaint survey November 2020.....	6
Wind Turbine Varied Operation And Amplitude Modulation	7

INTRODUCTION

DST Legal has commissioned L Huson & Associates Pty Ltd to review a number of reports prepared on the subject of the Lal Lal Wind Farm and to analyse data collected at property referenced as J14aa by SLR Consulting Australia Pty Ltd (SLR) in the course of a noise complaint investigation in the period 30 October 2020 to 1 December 2020.

The reports reviewed are:

Lal Lal Wind Farm Compliance Baseline Noise Monitoring SLR Ref No: 640.11872-R01-v1.1-20210216.docx February 2021;

Lal Lal Wind Farm - Elaine Post-construction Noise Assessment SLR Ref No: 640.11872-R04-v1.7-20220627.docx, 27 June 2022;

Lal Lal Wind Farm Noise Compliance Test Plan (NCTP) Marshall Day Acoustics (MDA) Rep. No. 003 R03 20170649 23 Jan 2018;

Lal Lal Wind Farm Verification of Post-construction Noise Assessment (Audit) Arup Australia Pty Ltd (Arup) Doc. Ref. 270849-00 22 June 2022;

Marshall Day Acoustics Lal Lal Noise Compliance Test Plan, Report No 003 R03 20170649, dated 23 January 2018 (NCTP).

Noise Logger Results, Dec 2020 prepared by SLR with accompanying Compliance Monitoring chart for J14aa covering 'All time' data.

Sound monitoring data collected by SLR was provided on a memory stick but no associated SCADA data was supplied for analysis.

DETAILED REVIEW

ARUP AUDIT

The Arup Audit is dated 22 June 2022 and refers to SLR report no. 640.11872-R04-v1.5, dated March 2022. However, this report had two later revisions (v1.6 and v1.7) that were not recorded in the audit.

The Audit concludes that:

“Compliance with the noise measurement and assessment methodology in the approved NCTP is considered to be consistent with compliance with the requirements of NZS 6808:2010, except where the NCTP requires alternative methods.”, and;

“The post-construction noise assessment (SLR Consulting Pty Ltd Lal Lal Wind Farm – Elaine, Compliance Noise Monitoring, Report No 640.11872-R04-v1.5, dated March 2022) demonstrates that the Elaine WEF complies with the noise limits set out in accordance with NZS 6808:2010 and Condition 23 of Planning Permit No PL-SP/05/0461/C.”

The Audit has flagged that the NCTP does not comply with NZS6808:2010. The Audit also confirms that only an earlier compliance noise monitoring report (v1.5) for the Elaine part of the Lal Lal Wind Farm was reviewed.

The body of the Audit report highlights the fact that the NCTP contains errors and notes that SLR in their compliance report (v1.5) have deviated from the valid data filtering process described in the NCTP but that such a deviation was acceptable in the Auditor's opinion.

Similarly, the Auditor has accepted the method of determining amplitude modulation described in the NCTP, even though the method does not comply with NZS6808:2010.

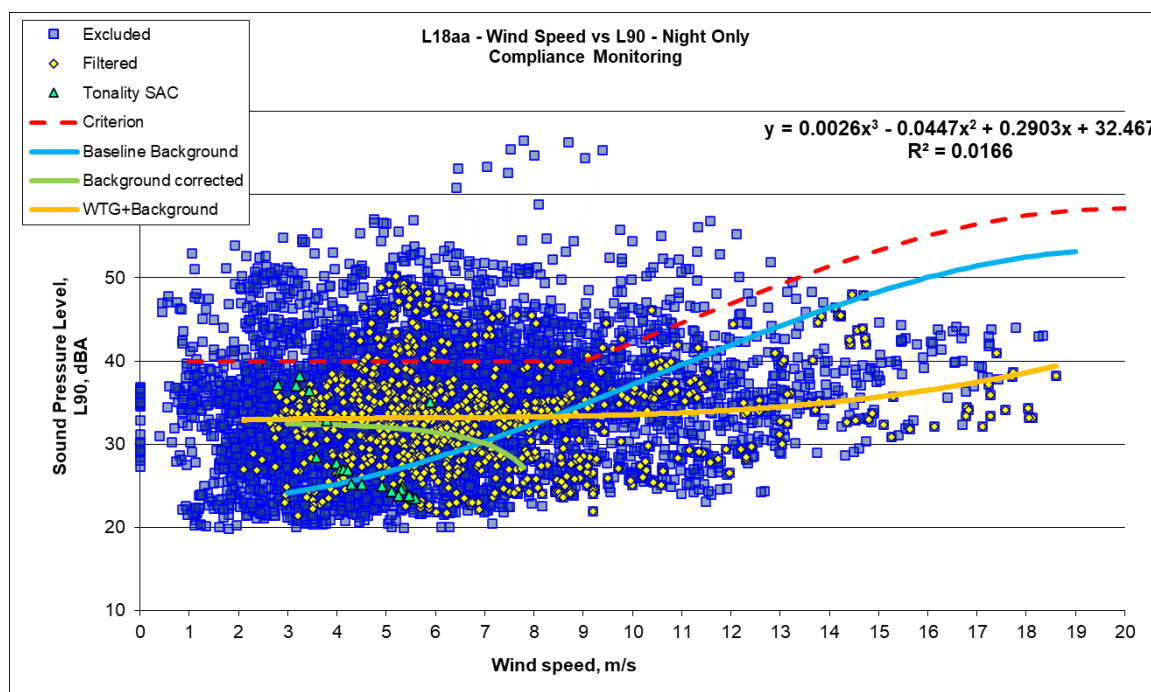
It is clear that the Auditor variously holds the standing of the NCTP higher than the requirement in the Permit to comply with NZS6808:2010 for amplitude modulation assessment, whilst at the same time acknowledging the NCTP has errors. Furthermore, the Auditor considers it acceptable to alter the NCTP in regard to data filtering.

The Audit has deemed the wind speed correction methods employed to be appropriate but fails to note that the latest Background noise report by SLR for property K15aa used wind speed data from the nacelle anemometer on wind turbine number E16 instead of data from a met mast.

The Audit incorrectly accepts that compliance monitoring analysis using corrections to a more remote, recently constructed met mast, was appropriate for use in assessing compliance at K15aa. This wind speed data mismatch error should have been identified in the Audit.

The Audit has accepted a noise measurement location approximately 85m from the K15aa dwelling, despite the NCTP and NZS6808:210 requiring Background and compliance measurements to be taken between 3.5m and 20m from a dwelling.

The Audit fails to recognise that post construction trend lines (Background noise plus wind farm noise) shown in the SLR compliance report for Elaine are regularly lower than the Baseline Background noise levels used to set target noise limits. An example from the SLR Elaine compliance report Figure 28 for dwelling L18aa is shown below:



Clearly, the Baseline Background trend line in this example, that is used to set target noise limits, is too high by a margin often in excess of 10 dB. Target noise limits should be 40 dB or Background plus 5 dB. So, assuming that Background alone must be lower than the 'Background plus wind farm' data then the target noise limit of 40 dB must extend at least up to about 14 m/s hub height wind speed, rather than 40 dB extending only to 9 m/s.

Accordingly, the Baseline Background noise levels used for compliance assessment, and which are also listed in the NCTP, are incorrect. This has significant implications for the Background correction method used and quoted compliance margin figures need to be reassessed.

ATTENDED SUBJECTIVE LISTENING SURVEYS

The SLR (Elaine) Post-construction Assessment Report includes a section on 'Attended subjective listening surveys' in Appendix D that also refers to surveys around the Yendon section of the Lal Lal Wind Farm.

All of the surveys near dwellings were during the day time (7am to 10pm) and many note that some wind turbines were not operating. Only survey locations 1, 2, 3 and 4 were attended at night (10pm to 7am) and then it was noted that at location 4 the 'Wind Farm seems to be below cut-in wind speed'.

AEMO data confirms that in the period from 5.30am to 7am at location 4 the Elaine section of the wind farm was generating no power, but the survey does note that a hum was discernible from the wind turbines during this time.

The attended measurements at locations 1, 2, 3, 5, 7 and 8 were all completed when wind farm generating capacity was less than 20%.

The SLR compliance report also notes that the hub height wind at K15aa on 11 June 2021 was 12.1 m/s @216° but also states that the wind was 'southerly' at 1-2 m/s. Other wind direction descriptor for observations near N31ab state there was a 'Northerly wind', yet hub height wind direction is tabled as 180°. These must be typographic errors.

Dwelling K15aa was observed 4 times on; 9 March 2021 (5%); 10 March 2021 (7%); 31 March 2021 (0%-no wind); 11 June 2021 (5%) with percentage generating capacity of the Elaine wind farm shown in brackets for each occasion. The operating conditions required for attended observations do not comply with the NCTP and the subjective assessment of SAC by SLR is unfounded.

Subjective notes also identify 'occasional brake release' and 'popping noise due to yaw brake'.

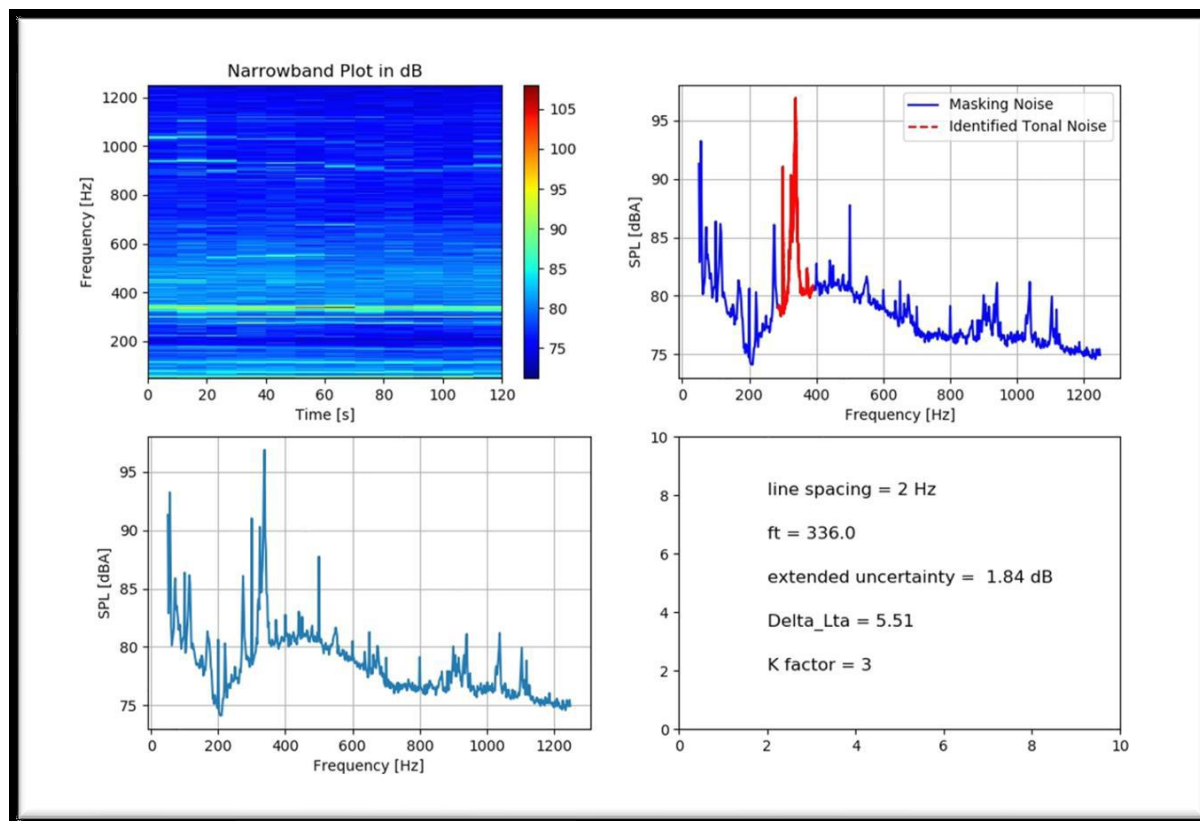
It may be that the Auditor does not understand what this refers to.

Before the Vestas V136 turbines yaw to face a changed wind direction a brake is released.

These brake release events sound like a loud 'clunk'. These impulsive sounds 'ring-down' with a frequency of approximately 340 Hz.

The transient brake release imparts a shock to the nacelle structure that excites local structural resonances and the resulting vibration is observed as sound. Indeed, the attended measurements regularly report audible hum around 340 Hz, 350 Hz and 360 Hz in Appendix D of the SLR

compliance report. Figure 4 of the SLR compliance report shows an example of tones that have been analysed, extracted below:



The attended surveys completed by SLR do not comply with the requirements of the NCTP and are deficient.

It is surprising that the Auditor considered the attended surveys to be acceptable and worthy of justification not to apply objective analysis of all special acoustic characteristics (SAC).

Overall, the Audit is considered superficial.

SLR COMPLAINT SURVEY NOVEMBER 2020

The noise logger results summary provided by SLR for the complaint investigation survey from 30 October 2021 to 1 December 2021 at four residential properties has the stated objective to check if the Lal Lal Wind Farm was compliant with permitted noise limits rather than investigate the cause of complaint.

A chart showing an ‘all time’ scatter chart was provided for J14aa and the raw recorded data was also supplied for this location with no chart for the night time period. Data and scatter charts for the other three residences surveyed were not provided.

The SLR summary notes that baseline background noise levels were not available for three out of the four residences surveyed, which included J14aa.

Baseline background noise result for K15aa was used to set target noise limits for J14aa.

The data gathered during the survey period throughout November 2020 cannot be used to assess compliance with permitted noise limits because the Lal Lal Wind Farm was not operating at full

capacity. However, data gathered during the survey can be informative for the purpose of investigating the noise complaints from residents at J14aa.

The SLR summary document of December 2020 did not have a night time chart provided with it. Subjective listening assessments were completed by SLR and although no detail has been provided about the observation times the summary report states that no SACs were observed at those times.

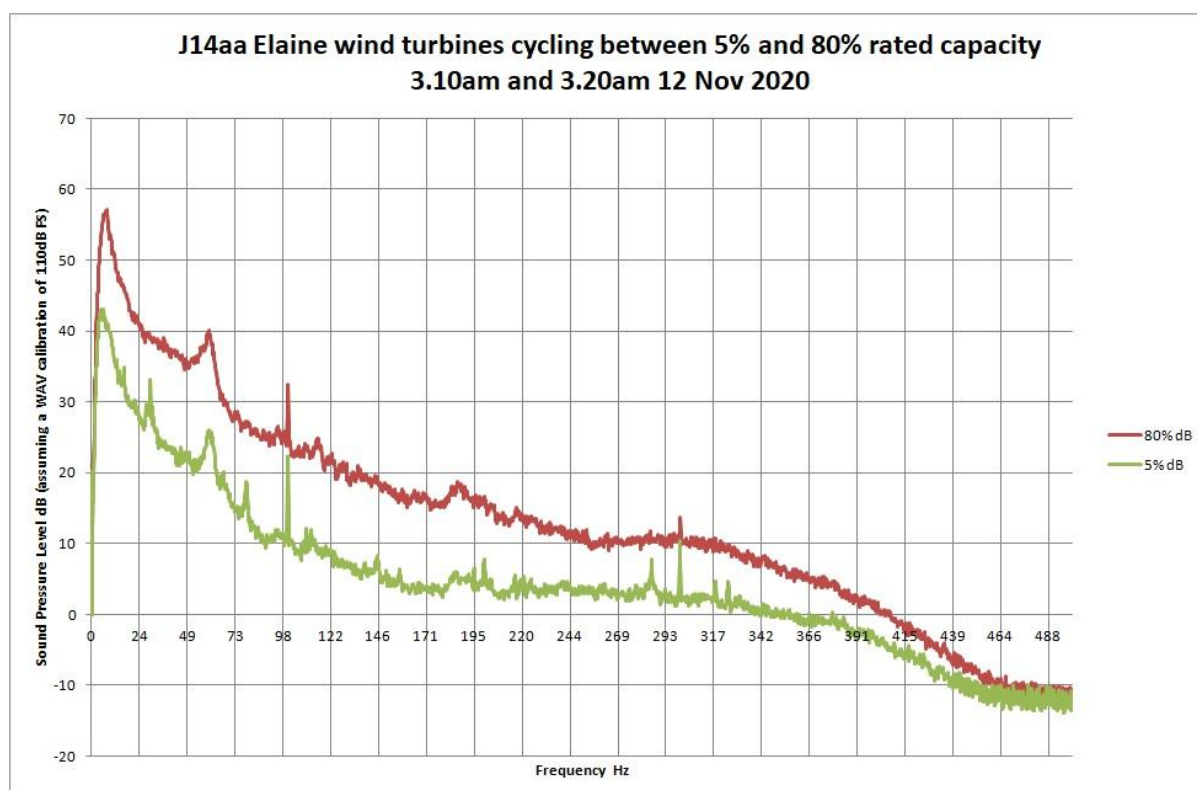
If observation times were limited to times of equipment deployment, maintenance and collection then the observation times would be 12 noon 30 October 2020 (20%); 9.30am 13 November 2020 (0%); 8.30am 1 December 2020 (60% to 70%) and the percentage of rated power produced by the Elaine wind farm section at the time are shown in brackets after each date. Observations made by SLR at these times are unrepresentative and do not comply with the NCTP.

On 1 December there were high winds that would have yielded 100% capacity so it is expected that some of the Elaine wind turbines were not operating.

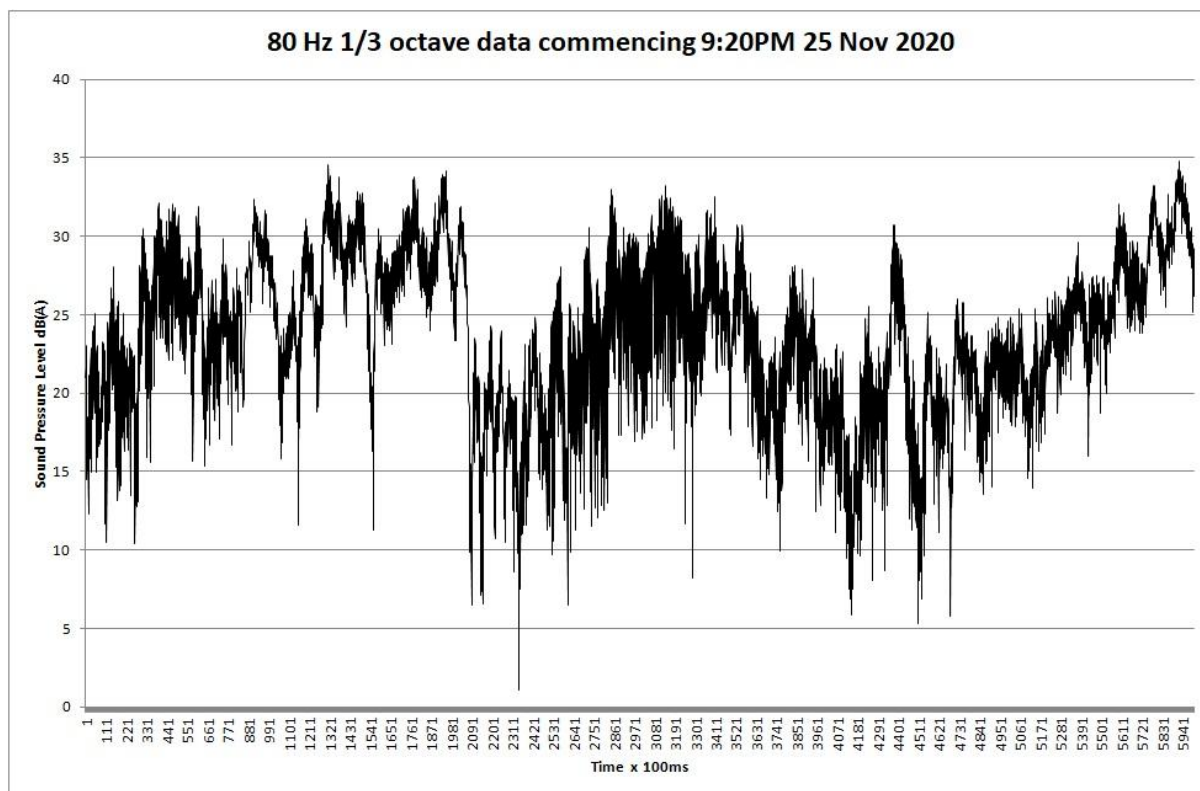
An analysis of the raw data provided by SLR has been completed to ascertain what may be the cause of noise complaints from residents at J14aa. A calibration tone could not be found on the WAV files provided so the calibration estimate of 110 dB full scale (FS) has been applied, which is based on comparing data recorded in 100ms samples during the survey.

WIND TURBINE VARIED OPERATION AND AMPLITUDE MODULATION

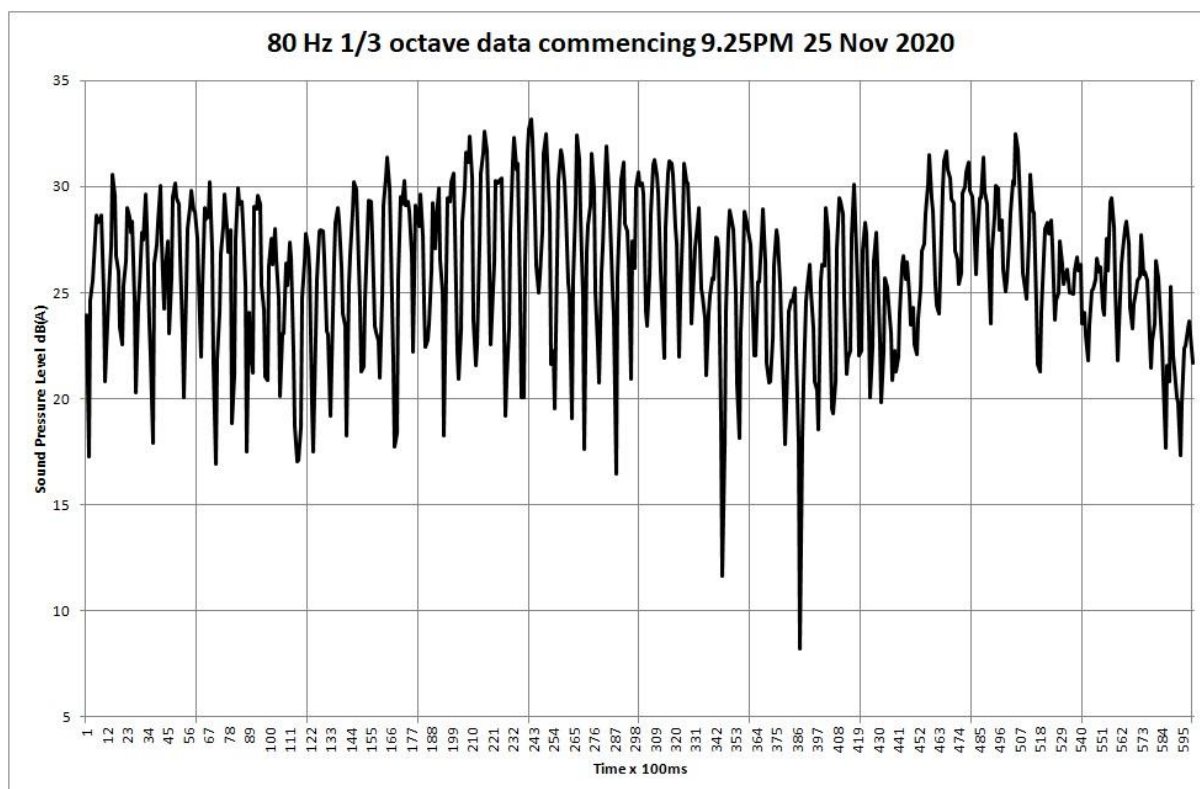
On 12 November 2020 the power generated by the Elaine wind farm cycled on and off a number of times between 3am and 4am. Spectrum analysis of 2-minute WAV data recorded at 3.10am (80% capacity) and 3.20am (5% capacity) show the difference in sound generated by the wind farm under different power generation but similar hub height wind speed in the chart below:



Listening tests accompanied by spectral analysis shows that the 80 Hz tone is at times modulated. For example, data analysed for 25 November 2020 shows the following 80 Hz 1/3 octave band data recorded at the rate of one sample every 100 ms:



A section of the chart above shows the amplitude modulation more clearly:



This is the analysis approach used in NZS6808:2010.

Note that there are a number of wind turbines to the north west of J14aa and each can contribute to cause independent modulation. The rotor can operate up to typically 14 rpm which equates to a blade pass frequency of 0.7 Hz.

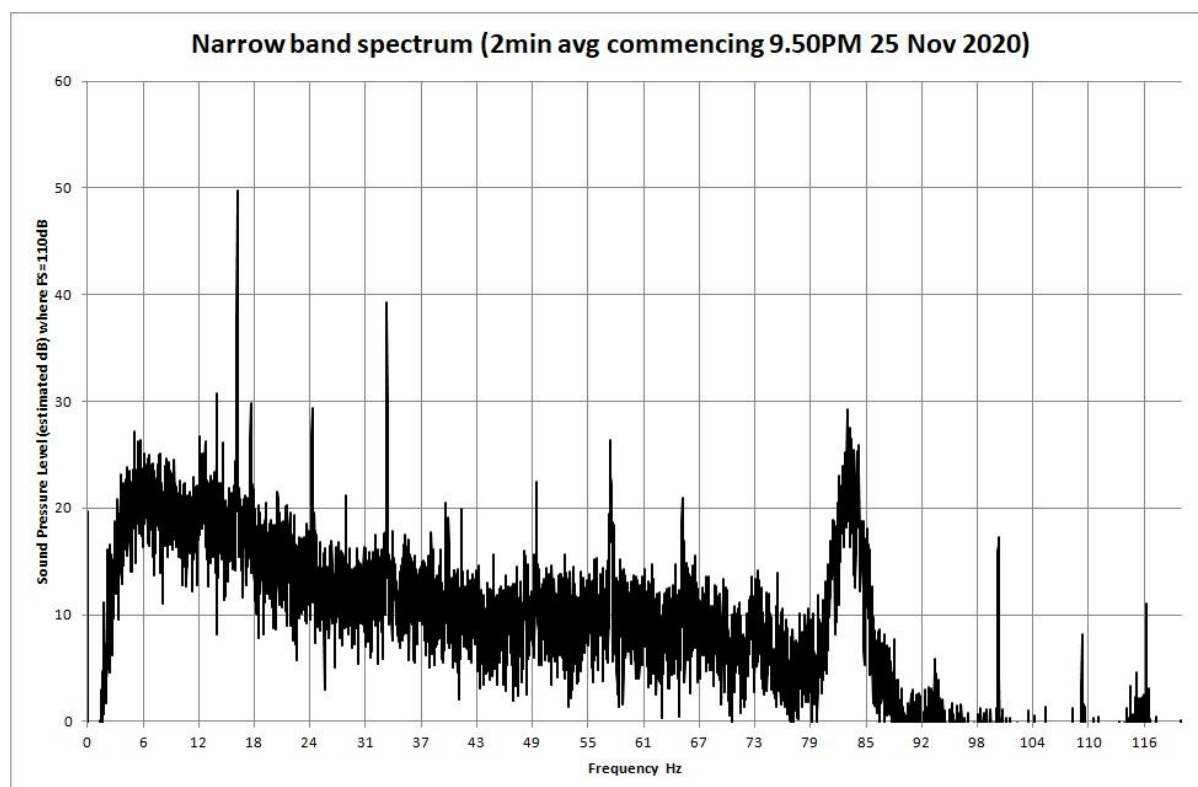
For a single wind turbine it is expected that modulation occurs at blade pass frequency (typically 0.7 Hz) and this is the modulation rate shown in the chart above. If two similar wind turbines operate out of synchronisation then the combined blade pass frequency of modulation will be observed at 1.4 Hz.

Amplitude modulation most often occurs at blade pass frequency from a wind turbine due to blade swish or for tones due to short term torque unloading of the gearbox as each blade passes the tower.

Such tonal modulation shows as side bands in the frequency spectrum that are spaced apart by the blade pass frequency. However, other mechanisms exist that can cause amplitude modulation of tones and the wind turbine manufacturers may provide advice in this regard for these particular wind turbines.

Amplitude modulation assessed in accordance with NZS6808:2010 would apply a penalty to the data that has not been considered by SLR and which the Arup Auditor has ignored.

Tones are observable from narrowband spectral analysis shown below, in a similar way that has been presented by way of example tonal analysis in Figure 4 of the SLR (Elaine) compliance assessment.



A suggestion has been made by legal representatives of the Lal Lal Wind Farm to the residents of J14aa that acoustic treatment of the dwelling may alleviate the noise distress that they have reported.

If such an option is to be explored then it is essential to know what the frequency content of wind farm noise is that needs to be blocked from entering the home.

Data provided by SLR when the Elaine wind farm was operating in constrained mode and when in full operation, as reported by SLR in their Post-construction Noise Assessment, show that sound energy to be considered in any acoustic treatment of the dwelling must attenuate both broadband low frequency sound and numerous low frequency tones below 500 Hz that dominate around 340 Hz, 80 Hz, 32 Hz, 25 Hz, and 16 Hz.

The generator within each wind turbine nacelle rotates at around 1500 rpm that equates to a frequency of 25 Hz, which is evident in spectral analysis of data analysed at J14aa.

SLR may have provided information described above to the Lal Lal Wind Farm as it is readily observable in data that SLR gathered.

Unfortunately, only a short summary document from SLR was provided to me with a single scatter chart for review. If a more comprehensive report is available from SLR to Lal Lal Wind Farm covering the survey they completed in November 2020 then a further review could be completed.

Prepared by



9 September 2022

W Les Huson BSc MSc CPhys MInstP MIOA MAAS